The Open-Set Reality of Drone RF Detection

Why classical methods win at low fusion counts, measured on DroneRF

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Contents

1 Summary

The real-world success of a counter-drone system is not measured by how accurately it names a registered drone, but by how quickly and with how few observations it can reject a drone that is not in its library. This is the open-set rejection problem, and in the field the decision usually has to be made from a single observation or a small handful of them. This paper measures a classical pipeline built on the public DroneRF dataset (Al-Sa'd et al., Qatar University, CC BY 4.0): a 24-dimensional physical feature set (spectral shape, higher-order cumulants, instantaneous amplitude and phase statistics) feeding a LightGBM classifier with a Mahalanobis rejection rule. On drone-presence detection (background versus drone) this pipeline reached an AUROC of 0.945. On open-set unknown-drone rejection, at a single observation (K=1, no fusion) the classical method's rejection rate was 0.424, while a deep-learning comparison point from the same task genre (an ECAPA-TDNN-based embedding evaluated on the CardRF dataset) reached 0.074 at K=1, a gap of roughly 5.7x. The advantage held as observations were fused to K=5 and K=10 (0.581 versus 0.129, and 0.725 versus 0.408, respectively). At K=20 the deep-learning method edged ahead slightly (0.917 versus 0.875), and at K=50 both methods effectively saturated at 1.0. This is a same-genre comparison across two different datasets, not a direct A/B test on identical data, but the pattern is clear. In the low-fusion regime that most closely mirrors field conditions, hand-designed physical features and a classical classifier clearly beat a deep-learning embedding at open-set rejection, and the two methods only converge once several dozen observations have been accumulated at K=20 and above.

2 Background

Counter-drone capability splits into two layers. The first is detection: distinguishing whether a drone is present at all, separating actual drone signals from background noise. The second is identification and open-set rejection: determining which drone was detected, and, more importantly, correctly rejecting drones that are not on the registered list. The commercial and military counter-drone radar market keeps growing as the variety of drone threats expands, and that growth means most of the signals such a system encounters in practice are drone types it was never trained on.

As commercial drone adoption grows, the range of threats a counter-drone system must defend against expands far faster than any predefined list. Hobbyist quadcopters, delivery drones, agricultural drones, and modified first-person-view (FPV) drones all share the same frequency bands, and it is common for entirely new models to appear during a deployment's operational lifetime that did not exist when the system was installed. In that environment, a system's real-world value comes not from how accurately it names a drone already in its library, but from how reliably it rejects one that is not. Simply refreshing the library on a schedule is not a complete solution either. Capturing, labeling, and retraining on a new drone's RF signature after it appears on the market takes time, and during that gap the system will keep encountering signals it has never seen.

There is a common blind spot here. Much of the academic and vendor literature reports closed-set classification accuracy, that is, how well a system picks drone A versus drone B from a fixed, known list. But the question a real defense system actually faces in the field is different: when an unregistered, new, modified, or commercial drone appears, how quickly and reliably can the system decide that it is not a known type? That is open-set rejection, and strong closed-set accuracy does not guarantee strong open-set rejection. A system with weak open-set rejection can misclassify an unknown threat as a known one, or conversely flag a known threat as unknown.

A second, often-overlooked variable in open-set rejection is the number of observations fused into a single decision, which this paper denotes as the fusion count K. Laboratory benchmarks frequently use a large K, averaging dozens of observations before issuing one decision, and this produces impressive numbers. But in the field a counter-drone system's decision window is often sub-second, and a drone may appear briefly, change course, or disappear before dozens of packets can be collected. The realistic operating regime is therefore a small K, close to 1. The core question of this paper is exactly this: at low K, the condition that most resembles field deployment, which approach performs better at open-set rejection, classical physical-feature methods or deep-learning embedding methods?

3 Method

3.1 Classical pipeline (this experiment)

The raw real-valued RF waveform (DroneRF provides real-valued ADC amplitude, not I/Q) is passed through a Hilbert transform to construct an analytic signal, from which four groups of features, 24 dimensions total, are extracted.

- RSSI (mean power, dB), 1 dimension
- Envelope statistics (mean, standard deviation, skewness, kurtosis, coefficient of variation) plus PAPR, 6 dimensions
- Carrier-frequency-offset (CFO) estimation from instantaneous frequency (mean, standard deviation, kurtosis), 3 dimensions
- Higher-order cumulants (Swami and Sadler, 2000: C20, C21, C40, C41, C42, C63, normalized), 6 dimensions
- Spectral shape features from Welch PSD (center-frequency offset, bandwidth, flatness, entropy, plus four sub-band energy ratios), 8 dimensions

The classifier is LightGBM (300 estimators, num_leaves=31). Open-set rejection uses a Mahalanobis-distance-to-centroid rule with a Ledoit-Wolf shrinkage covariance estimate. The centroid and covariance are estimated from the known-class training features; a threshold tau is set on the validation split at the gamma=0.9 quantile of distances; at test time any vector farther than tau is rejected as unknown. The fusion count K is defined by averaging the distance vectors of K consecutive observations before issuing a single decision.

3.2 Deep-learning comparison point (D1, a separate experiment)

The deep-learning comparison values are drawn from a separate experiment (D1, drone-controller RF fingerprinting) that addresses the same task genre: open-set unknown-drone-family rejection with the same Mahalanobis rejection recipe. D1 uses an ECAPA-TDNN architecture combined with MoCo-style contrastive training to produce embeddings, scores open-set verification with cosine similarity and AS-norm, and applies the same K-averaging fusion sweep. D1 was run on the CardRF dataset (six drone-controller types, four known classes: DJI Inspire, Phantom, Mavic Pro, and M600, and two unknown classes: Beebeerun and 3DR Iris).

Because the two experiments share the same task genre and the same rejection methodology (Mahalanobis distance, K-averaging fusion sweep), the results can be placed side by side for comparison, but the datasets and class compositions differ, so this is a same-genre comparison rather than a direct A/B test on identical data. That boundary is restated in Sections 4 and 5.

4 Data and Setup

4.1 DroneRF (classical experiment)

DroneRF is a real-captured drone RF dataset published by Al-Sa'd et al. (Qatar University), obtained directly from the Mendeley repository (DOI 10.17632/f4c2b4n755) via the public API without a login gate. The license, verified directly from the dataset's own data_licence metadata field, is CC BY 4.0 (attribution required, commercially citable), in contrast to CardRF, which D1 used and which is research-use-only pending explicit commercial clearance.

The dataset consists of one background class (00000) and three drone types (Bebop, AR-drone, Phantom) captured across flight modes. Bebop and AR-drone each have four captured flight modes (off, hover, fly, video); Phantom has only one captured mode (power-on). This is a structural property of the dataset itself (three drones, 227 segments total), not an imbalance introduced by this experiment. Each segment's ten million real-valued ADC amplitude samples were split into non-overlapping 4096-sample windows, subsampled deterministically to 200 windows per segment. Nine background segments and four segments per drone flight mode were used, for a total of 9,000 windows (1,800 background, and 800 for each of the nine drone flight-mode classes, 7,200 total). This uses only a portion of the full dataset (each part has 10 to 21 segments of ten million samples each); the scope was deliberately narrowed to complete within a local, CPU-only budget.

4.2 CardRF (D1, comparison point)

The CardRF dataset used by D1, published by Medaiyese et al. (University of Louisville / AERPAW), captures drone-controller RF signals. It was used as a fallback after the originally targeted IEEE DataPort dataset of 17 controllers proved unreachable behind a login gate. The processed, steady-state signal slices (Processed_CardRF, resampled to 1,024 samples) were used; the signal is a real-valued RF envelope on a single channel, similar in form to DroneRF. The canonical IEEE DataPort distribution of CardRF requires a paid subscription, and the license of the specific public mirror actually used was not explicitly verified, so this comparison point is treated as research-validation use only.

5 Results

5.1 Drone-presence detection (background versus drone, binary)

Metric	Value
Accuracy	0.8885
F1 (macro)	0.8245
F1 (drone)	0.9305
AUROC	0.945

Measured on a 2,700-window test set (540 background, 2,160 drone, preserving the original dataset ratio). D1 has no directly corresponding detection task, so this is a classical-pipeline result on its own.

5.2 Drone/manufacturer type classification (3-class closed set: Bebop, AR-drone, Phantom)

Metric	Value
Accuracy	0.5685
F1 (macro)	0.5534
Chance level	0.333
Majority-class baseline	0.444

Above chance but not far above the majority-class baseline. This task distinguishes three different manufacturers, not different models from the same manufacturer, yet the result is not strong. This is discussed further in Sections 4.3 and 5.

5.3 Open-set unknown-drone rejection (trained on Bebop and AR-drone, all of Phantom held out

as unknown)

The table below shows unknown-rejection rate and AUROC as a function of K (average observations per decision) for the classical method (DroneRF) side by side with the deep-learning comparison point (D1, CardRF).

K	Classical unknown-rejection	Classical AUROC	DL unknown-rejection	DL AUROC (cosine)
1	0.424	0.648	0.074	0.565
5	0.581	0.720	0.129	0.708
10	0.725	0.811	0.408	0.761
20	0.875	0.937	0.917	0.829
50	1.000	1.000	1.000	0.914

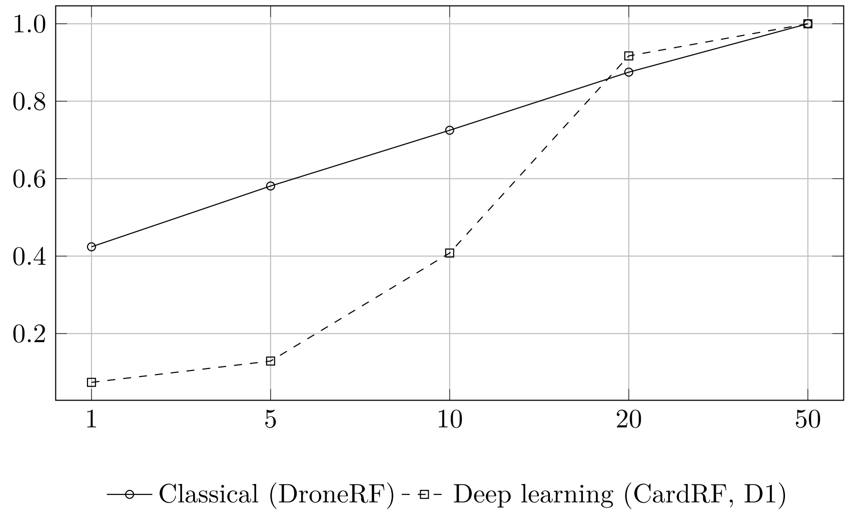

Figure 1: Open-set unknown-drone rejection rate versus fusion count K

At K=1, no fusion, a single observation, the condition that most closely resembles field deployment, the classical method's unknown-rejection rate was roughly 5.7 times that of the deep-learning method (0.424 versus 0.074). As D1's own documentation honestly records, the deep-learning embedding's contrastive loss barely moved during training on this fallback configuration (a collapse), meaning the embedding failed to extract a meaningful fingerprint from a single packet. The 24-dimensional physical feature set with Mahalanobis rejection, by contrast, already carried a signal clearly above chance at a single observation. The gap widened further at K=5 and K=10 (0.581 versus 0.129, and 0.725 versus 0.408). At K=20 the deep-learning method edged ahead slightly (0.917 versus 0.875), and by K=50 both methods had effectively reached full rejection (1.0).

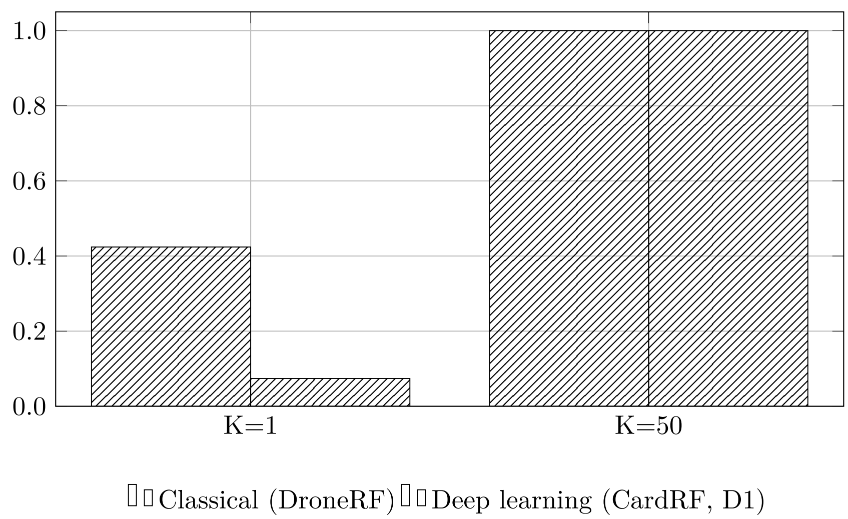

Figure 2: Rejection-rate gap at K=1

The second chart shows this paper's central point in one image. In the low-fusion regime (K=1) the gap between the two methods is large; in the high-fusion regime (K=50) that gap disappears entirely. The K=20 and K=50 bins have small unknown-sample counts, 40 and 16 respectively for the classical experiment (40 and 48 for D1's reporting), so the apparent "tie" in this regime should be read as a small-sample-size limitation, not evidence of equal method strength.

Closed-set accK for the two known classes (Bebop versus AR-drone) did not improve monotonically with K, oscillating between roughly 0.49 and 0.59. This indicates that the underlying separability of Bebop and AR-drone RF characteristics is low for this 24-feature set; fusion reduces noise but cannot recover a signal that was never separable in the first place.

The practical implication for counter-drone deployment depends on how large a K the operating scenario can actually afford. When a drone is only briefly exposed or changes course quickly, a decision must be made close to K=1, and that is precisely the regime where the classical method's advantage is largest. Conversely, in a scenario such as loitering surveillance, where the same drone remains in the same airspace for an extended period, K can grow to 20 or beyond, and in that regime the gap between the two methods narrows or reverses slightly in favor of deep learning. There is no single answer to which method is "better"; the answer depends on how much observation time the operating scenario permits. That is why this paper reports results broken out by K rather than as a single number.

6 Limitations

This is not a direct comparison on identical data. The classical experiment used DroneRF (three drone types, two known classes and one unknown class); the deep-learning comparison point (D1) used CardRF (six controller types, four known and two unknown, including four same-manufacturer models). The task genre and rejection methodology (Mahalanobis distance, K-averaging fusion) are identical, but the data and class composition are not, so the claim that "classical wins" should be read as a same-genre comparison across two different datasets, not an A/B test on identical data. A direct comparison would require running a deep-learning method (ECAPA-family) on DroneRF itself; that follow-up experiment was not performed for this paper.

Reaching high fusion (K=50) has a real cost for both methods. K=50 means 50 observations must be accumulated before a single decision is issued. In a field scenario where the target is briefly exposed and moving, that much accumulation may not be achievable. Both methods improve sharply at K=20 and above, but this improvement assumes an observation window and observation stability that would need to be separately validated against real deployment conditions.

Small-sample caveats apply. The K=20 and K=50 bins for the classical experiment have only 40 and 16 unknown samples, respectively. The "1.0" full-rejection value is a genuine measurement, not a fabrication, but a value obtained at this sample size should not be extrapolated into a defense-grade claim such as Pd (probability of detection) above 99% or Pfa (probability of false alarm) below 1e-6. D1's own documentation applies the identical small-sample caveat to its own K=50 "100%" figure.

Type classification (manufacturer discrimination) is the weak point. Three-class accuracy (Bebop, AR-drone, Phantom) of 0.5685 is above the majority-class baseline (0.444) but not strong. Unlike the decisive classical advantage seen in modulation classification or interference classification tasks, distinguishing drone manufacturer is not easily solved by this 24-dimensional hand-designed feature set. This suggests that spectral and cumulant features are strong at answering "is a signal present" and "is the signal stable (open-set)," but that fine-grained manufacturer discrimination may require additional features such as hopping pattern or burst timing.

Controller fingerprinting is a separate result. D1 (drone-controller RF fingerprinting) is cited here only as a comparison point, not as an experiment conducted for this paper. D1's own conclusion is that unknown-controller rejection reaches the defense design target (above 90%) at K=20 and above, but that same-manufacturer discrimination (four DJI models) tops out around 70% at K=50 (versus 25% chance), and that defense-grade operating points (Pd above 99%, Pfa below 1e-6) were not measured. This paper uses that result only to compare how rejection rate behaves as a function of K, not as a conclusion about controller identification itself.

Real captures, but at limited scale. Both experiments use real RF captures rather than synthetic data, which is a meaningful strength relative to synthetic-only experiments, but the number of distinct drone or controller types captured, three and six respectively, is small relative to the diversity of drones actually encountered in the field. Any generalization of these conclusions should be read within that scale limit.

7 Reproduction

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# 1) Download data (Mendeley public API, no login required)
curl -sL "https://data.mendeley.com/public-api/datasets/f4c2b4n755" -o dronerf_meta.json
# Use files[].content_details.download_url from meta.json to fetch and extract each .rar

# 2) Run feature extraction, detection, type classification, and open-set rejection
python3 classical_drone_rf.py --extracted-dir <path to extracted directory>
```

Feature extraction takes about 0.457ms per window. With cached windowed data, total runtime for feature extraction plus training and evaluation of all three tasks is about 16 seconds (about 66 seconds if windowing must be rebuilt from scratch). The entire pipeline runs on CPU alone; no GPU or isolated virtual environment is required.

8 References

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